SUPPLEMENTARY MATERIAL

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Table 7: Comparison of deepfake video dataset and its forgery techniques.

Dataset	Release Year	Manipulation Techniques
UADFV [27]	2018	FacesSwap
DeepfakeTIMIT [16]	2018	Faceswap-GAN
FF++ [24]	2019	[c]FacesSwap, Face2Face,
DeepFakes, Neural Textures		
Google DFD [24]	2019	FaceSwap, Face2Face
Celeb-DF [18]	2020	Improved DeepFake
DeeperForensics [14]	2020	DFAE, StyleGAN
DFDC [8]	2020	[c]DFAE, MM/NN,
Neural Talking Heads, FSGAN,		
StyleGAN, TTS Skins		
KoDF [17]	2021	[c]FaceSwap, LipGAN,
DeepFaceLab		
FakeAVCeleb [15]	2021	[c]FSGAN, FaceSwap,
Wav2Lip, SV2TTS		
DeepSpeak [3]	2024	[c]CodeFormer GAN, FaceFusion,
FaceFusion Live		
DigiFakeAV (ours)	2025	[c]V-Express, Hallo,
Hallo2, Sonic,		
Echomimic, CosyVoice2		

A. MORE DETAILS OF DIGIFAKEAV

A.1. Comparisons with Previous Datasets

We investigate the generation methods of the first, second, and third-generation datasets, demonstrating that their forgery techniques converge and have become outdated. In contrast, our approach leverages the latest digital human forgery technologies, which significantly advance beyond traditional face-swapping and GAN-based frameworks. As shown in Tab.7, by integrating cutting-edge models such as V-Express [25], Hallo [26], Sonic [13], and CosyVoice2 [10], we achieve high-fidelity and dynamic digital humans that convincingly replicate multi-modal features including facial expressions, lip synchronization, and voice characteristics.

Our investigation reveals that digital human forgery techniques have already yielded a lot of impressive results and continue to develop rapidly. Building on these insights, we further validate the inherent challenges posed by digital-human forgery through our experimental findings, and from this foundation, we have constructed the DigiFakeAV dataset to reflect the observed complexities and promote future detection research.

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A.2. Diffusion-based Methods for Data Synthesis

As demonstrated in Tab.9, we conduct a comprehensive survey and extensive experiments on a diverse collection of state-of-the-art (SOTA) diffusion-based digital human generation methods. These approaches exhibit outstanding audio–visual conditional generation capabilities in architectural design and modality-control strategies. The brief introduction is as follows:

•**V-Express [25]**. By progressively dropping strong signals, the proposed strategy effectively balances multimodal conditions, enabling weak signals to gradually participate in control, thereby achieving coordinated pose, image, and audio-driven generation of highly synchronized deepfake videos.

• **Sonic [13]**. Sonic employs time-aware fusion across video segments to disentangle head motion from facial expressions. By leveraging long-range audio context, it achieves continuous and natural motion synthesis, enabling the generation of coherent and realistic deepfake videos.

• **EchoMimic [4]**. By integrating hybrid conditioning of audio and facial keypoints, and employing regional mapping alongside motion synchronization strategies, this method generates deepfake videos featuring highly natural facial expressions and stable audio-visual synchronization.

• **Hallo [26]**. A latent diffusion model with hierarchical audio-driven visual synthesis is employed, combined with a multi-level cross-attention mechanism to achieve finer audio-visual alignment, thereby enhancing the continuity and realism of the synthesized videos.

• **Hallo2 [6]**. Hallo2 is optimized for long-duration and ultra-high-resolution video generation. Employing vector quantization (VQ)-based encoding combined with a temporal alignment mechanism, it enables stable synthesis of long-form deepfake videos.

• **CosyVoice 2 [10]** . As shown in Table 8, we select CosyVoice 2 as our default speech synthesis system due to its superior performance. Specifically, CosyVoice 2 achieves the lowest word error rate (WER), the highest naturalness mean opinion score (NMOS), and strong speaker similarity (SS=0.745), outperforming other TTS models on all key metrics. These advantages ensure high-quality and consistent audio for our experiments. By employing multi-stage semantic decoding and conditional flow matching techniques, along

Table 8: Comparison of content consistency (WER), speaker similarity (SS), and speech quality (NMOS) of various speech synthesis models on the LibriSpeech test-clean subset. CosyVoice 2 achieves the lowest WER, highest NMOS, and best speaker similarity (SS = 0.749), outperforming other baselines.

Model	WER (%) ↓	NMOS ↑	SS ↑
Human	2.66	3.84	0.697
ChatTTS [1]	6.84	3.89	–
GPT-SoVITs [23]	5.13	3.93	0.405
OpenVoice [22]	3.47	3.87	0.299
ParlerTTS [19]	3.16	3.86	–
EmotiVoice [28]	3.14	3.93	–
CosyVoice [9]	2.89	3.93	0.743
CosyVoice 2	**2.47**	**3.96**	**0.749**

with a unified streaming and non-streaming language model, this approach provides an efficient and stable foundation for speech synthesis in audio-driven digital human videos.

A.3. Quality Standards for Data Synthesis

To construct a high-quality forgery dataset, we establish a series of rigorous quality control standards encompassing multiple widely recognized evaluation metrics. Specifically, we adopt Fréchet Inception Distance (FID) to assess the visual realism of each generated sample, and Synchronization Consistency (Sync-C) to evaluate audio-visual synchronization. Furthermore, additional quantitative metrics, including Fréchet Video Distance (FVD), synchronization difference (Sync-D), extended FID (E-FID), feature similarity (F-SIM), smoothness (Smooth), and runtime, are considered to comprehensively reflect the performance of generative models from multiple perspectives.

All generated samples are automatically evaluated using the aforementioned metrics. During the quality control process, hard thresholds are set for FID and Sync-C: samples with FID scores higher than 45 (often indicating noticeable visual distortion) or Sync-C scores lower than 1.0 (indicating severe audio-visual misalignment) were automatically discarded. The other indicators, such as FVD, E-FID, F-SIM, and Smooth, functioned as auxiliary references to further analyze the temporal consistency, semantic accuracy, and perceptual quality across different generative methods, ultimately guiding the selection of high-quality clips. In addition, all videos that passed automatic filtering undergo a further manual review, which specifically focuses on subtle artifacts and semantic errors that might not be reflected by quantitative metrics, ensuring samples meet the expected quality standards in both objective and subjective aspects.

The final selected samples exhibit excellent performance across all evaluated metrics, with representative results shown in Tab. 10. By employing a multi-metric and multi-stage quality control pipeline, we significantly improve the overall quality and generalization of our dataset.

B. MORE DETAILS OF DEEPFAKE DETECTION METHODS

Overview of the deepfake detection methods used in our experimental evaluation. These models represent a diverse range of architectures and forgery cues, including spatial artifacts, temporal inconsistencies, frequency-domain anomalies, and audio-visual synchronization signals. We describe each method's core design principles and its relevance to detecting various types of video forgeries present in datasets such as DigiFakeAV.

- **Meso4 [2].** It is a lightweight convolutional neural network architecture designed for efficient extraction of facial forgery features, facilitating rapid deepfake detection.
- **MesoInception4 [2].** It is an extension of Meso4 that integrates Inception modules to enhance multi-scale feature representation and capture more discriminative cues.
- **Xception-c23 [24].** It is a deepfake detection approach based on the Xception architecture, leveraging depthwise separable convolutions for improved feature extraction and model efficiency.
- **Capsule [20].** It is a classification method employing capsule networks to preserve hierarchical pose relationships and improve robustness against spatial transformations.
- **HeadPose [27].** It is a technique that utilizes head pose estimation as discriminative features for classification between genuine and forged videos.
- **F3-Net [21].** It is a method that discriminates forged videos by extracting statistical features from the local frequency domain, effectively capturing subtle forgery artifacts.
- **Cross Efficient ViT [5].** It is an approach that incorporates ViT architectures augmented with a voting mechanism to robustly identify deepfake content.
- **SFIConv [12].** It is a method to combine information from the spatial and frequency domains using a modified convolutional backbone network to enhance the detection capabilities of counterfeit features.
- **SSVF [11].** It is a framework that exploits the intrinsic synchronization between visual and audio streams, and employs autoregressive models to learn temporal dependencies characteristic of authentic videos and detect anomalies effectively.

C. MORE DETAILS OF IMPLEMENTATION

C.1. Preprocessing

To ensure consistency and precision in the facial inputs used for downstream feature extraction and model training, we adopt RetinaFace [7], a deep learning–based face detector and aligner. Our preprocessing pipeline comprises three sequential steps:

Table 9: Comparison of audio-driven diffusion-based digital human generation methods.

Algorithm	Team	PDF	GitHub	OSS	Train Data	Test Data	Model	Precision	Advantages	Limitations
EMO	Alibaba Group	2402.17485	—	none	HDTF, VFHQ	HDTF (10%)	SD1.5, 453 M, 1.7 GB	fp16	Large motion amplitude	Jittery head rotation; weak control
EchoMimic	Ant Group	2407.08136	repo	infer+model	HDTF, CelebV-HQ	HDTF (10%), CelebV-HQ (10%)	SD1.5, 453 M, 1.7 GB	fp16	Natural expressions	Color shifts; ID mismatch
V-Express	Nanjing Univ. / Tencent AI Lab	2406.02511	repo	train+infer+model	HDTF	HDTF (10%), CelebV-HQ (10%)	SD1.5, 453 M, 1.7 GB	fp16	Realistic video	Needs extra KPS; color shifts
Hallo	Fudan / Baidu	2406.08801	repo	train+infer+model	VFHQ, HDTF, CelebV-HQ	CelebV-HQ	SD1.5, 453 M, 1.7 GB	fp16	Stable appearance; fewer artifacts	Small motion; occasional ghosting
Hallo2	Fudan / Baidu	2410.07718	repo	train+infer+model	VFHQ, HDTF, CelebV-HQ	CCv2, HDTF	SD1.5, 453 M, 1.7 GB	fp16	Multi-mode support	May harm ID; occasional errors
Sonic	Tencent	2411.16331	repo	infer+model	VFHQ, CelebV-Text, VoxCeleb2	CCv2, HDTF	stable-video-diffusion-xt-1-1	fp16	Better AV sync	Less detailed facial control

Table 10: Quantitative comparison with state-of-the-art methods on the HDTF and CelebV-HQ test sets. **Bolded** values indicate the best performance.

Dataset	Method	FID↓	FVD↓	Sync-C↑	Sync-D↓	E-FID↓	F-SIM↑	Smooth↑	Runtime
HDTF	V-Express (2024)	40.776	758.023	1.826	12.594	2.364	0.9598	0.9954	39.65
	Hallo (2024)	38.773	342.548	4.136	9.445	1.203	0.9566	0.9974	45.75
	Hallo2 (2024)	35.206	318.436	**4.154**	9.017	**1.170**	0.9567	**0.9976**	57.04
	EchoMimic (2024)	38.077	327.110	3.171	9.435	1.244	0.9505	0.9969	17.85
	Sonic (2024)	**29.104**	**301.173**	4.197	**9.371**	1.245	**0.9592**	0.9974	17.04
CelebV-HQ	V-Express (2024)	65.400	889.209	2.809	13.253	2.713	0.9597	0.9967	39.65
	Hallo (2024)	52.305	416.492	3.495	10.140	2.192	0.9572	0.9972	45.75
	Hallo2 (2024)	48.436	398.138	**3.580**	9.796	**1.696**	0.9593	**0.9979**	57.04
	EchoMimic (2024)	48.267	569.870	1.249	10.754	2.136	0.9562	0.9962	5.45
	Sonic (2024)	**43.137**	**483.108**	2.689	**10.194**	1.783	**0.9624**	0.9972	17.04

Face Detection. We apply RetinaFace to each input frame to obtain a tight face bounding box together with five facial landmarks (the inner and outer eye corners, nose tip, and mouth corners), thus providing accurate localization of key facial regions.

Geometric Alignment. Using the detected five landmarks, we compute a similarity transformation matrix that normalizes each face to a canonical pose (e.g., horizontally level eyes and fixed relative positions of the nose tip and mouth corners).

Image Cropping. In the aligned coordinate system, we crop a centered, fixed-scale facial region from the original image and resize it to 224 × 224 pixels.

By applying these implementations, we produce high-quality, uniformly aligned face images that serve as reliable and stable inputs for subsequent audio-driven portrait synthesis or deepfake detection tasks.

C.2. Training Settings

To fully exploit the correspondence between the visual and audio modalities, we employ a 3D ResNet as our video encoder and Xception as our audio encoder, projecting both into a common 1024-dimensional feature space. Input video frames are resized to 224×224 pixels.

The entire model is trained end-to-end in PyTorch using stochastic gradient descent (SGD) with momentum 0.9 and weight decay 1×10^{-4}. We set the initial learning rate to 0.01 and apply a final learning-rate scaling factor (`lrf`) of 0.01. We train the model for 50 epochs with a batch size of 16, using two NVIDIA A100 GPUs.

D. MORE DETAILS OF BASELINE COMPARISON

In this section, we provide a more detailed evaluation of the previously established three generations of datasets, as illustrated in Tab.11. Through comprehensive experiments, we further demonstrate the challenging nature of our proposed dataset.

The performance of state-of-the-art deepfake detection methods on DigiFakeAV reveals significant limitations in current techniques when confronting diffusion-based digital human forgeries. As shown in Table 11 and Fig. 5, most models exhibit marked performance degradation compared to previous benchmarks, with AUC scores dropping sharply across categories. Naive CNN-based approaches, such as Meso4 and MesoInception4, performed near chance level (50.1–63.8% AUC), underscoring their inability to detect subtle artifacts introduced by advanced forgery techniques. In contrast, *spatial-based methods* like Capsule achieved moderate results (65.3% AUC), while frequency-domain methods (e.g., F3-Net at 66.4% AUC) demonstrated partial efficacy, both significantly below their performance on earlier datasets.

The most striking decline occurred in temporal methods: Cross Efficient ViT, which achieved 99.1% AUC on DFDC, dropped to 52.2% on DigiFakeAV. This suggests that diffusion-generated videos exhibit enhanced temporal coherence, a critical vulnerability for detectors relying on motion inconsistencies. Hybrid-domain methods like SFIConv (71.2% AUC) outperformed others by combining spectral and spatial features, yet this remains far below its near-perfect results on DF-TIMIT (100%) and FakeAVCeleb (93.0%). The most striking failure, however, was observed in multimodal detectors: SSVF's AUC plummeted from 94.5% on FakeAVCeleb to 51.0%, indicating that diffusion models have effectively minimized audio-visual misalignments, a key detection cue for cross-modal approaches.

These findings collectively emphasize that existing strategies relying on conventional feature fusion or domain-specific priors are insufficient for detecting modern diffusion-based forgeries. Novel approaches must address the unique challenges posed by high-fidelity spatiotemporal and cross-modal consistency in synthetic videos.

E. MORE DETAILS OF ABLATION STUDY

To validate the effectiveness of our audio-visual spatiotemporal fusion strategy, we conduct an ablation study on the DigiFakeAV dataset. The results are summarized as follows: (1) **Baseline Model**: the model uses only cross-entropy loss (L_{ce}) without any attention mechanisms, achieves an AUC of 73.6. (2) **With Contrastive Loss**: the model adds contrastive loss (L_{con}), improves the AUC to 77.4, demonstrating the benefit of enhancing feature alignment across modalities. (3) **Full Model**: the model incorporates both cross-attention (CrossAtt) and self-attention (SelfAtt), further boosting performance to an AUC of 80.1, highlighting the synergistic effect of these components.

Through our experiments, we obtain some key observations: (1) **Cross-Attention:** the cross-attention module plays a critical role in integrating complementary cues across modalities, enabling mutual guidance between visual and audio features. This is essential for detecting subtle digital human artifacts. (2) **Self-Attention:** the self-attention significantly enhances temporal coherence within each modality, contributing to improved robustness against complex forgery techniques.

The ablation study confirms that both inter-modal and intra-modal attention mechanisms are crucial for detecting advanced deepfake techniques, such as those generated using diffusion models. These findings support the design choices made in our model architecture.

Table 11: AUC scores (%) of various methods on previous datasets and our DigiFakeAV

Type	Method	UADFV	DF-TIMIT		FF-DF	DFD	DFDC	Celeb-DF	FakeAVCeleb	DigiFakeAV
			LQ	HQ						
Naive	Meso4	84.3	87.8	68.4	84.7	76.0	75.3	54.8	60.9	50.1
Naive	MesoInception4	82.1	80.4	62.7	83.0	75.9	73.2	53.6	61.7	63.8
Naive	Xception-c23	91.2	95.9	94.4	99.7	85.9	72.2	65.3	72.5	66.1
Spatial	Capsule	61.3	78.4	74.4	96.6	64.0	53.3	57.5	70.9	65.3
Spatial	HeadPose	89.0	55.1	53.2	47.3	56.1	55.9	54.6	49.0	51.7
Frequency	F3-Net	51.1	99.8	99.4	93.7	79.8	88.4	95.9	91.3	66.4
Temporal	Cross Efficient ViT	94.4	50.4	55.8	99.1	93.0	95.1	86.7	80.5	52.2
Multi model	SSVF	-	-	-	-	-	-	-	94.5	51.0
Hybrid Domain	SFIConv	54.4	100.0	100.0	95.9	88.0	96.7	95.8	93.0	71.2

F. ETHICS AND USAGE GUIDELINES OF DIGIFAKEAV

F.1. Ethical Review

All data generation and usage processes in this study strictly adhere to ethical guidelines. All real video data are sourced from publicly available authorized datasets (e.g., HDTF and CelebV-HQ) and comply with relevant copyright and privacy terms. The synthetic data generation process ensures no risk of violating the real individuals' portrait rights or privacy. Additionally, the DigiFakeAV dataset is limited to academic research purposes and must not be used for commercial or malicious purposes without permission. Using the dataset to generate or disseminate illegally forged content is explicitly prohibited.

F.2. Social Impact

The detection technology in this study provides a critical tool for curbing the spread of deepfake content. As forged videos generated by diffusion models achieve breakthroughs in visual realism and cross-modal consistency (e.g., the 68% misjudgment rate among users in DigiFakeAV indicates), their threats to social trust and public safety have become increasingly severe. Our detection method (DigiShield), by capturing spatiotemporal inconsistencies and cross-modal discrepancies, offers an effective means to identify such advanced forgeries, assisting social media platforms and law enforcement in rapidly responding to misinformation.

However, the study acknowledges potential risks of abuse. For example, the dataset might be used to improve forgery technologies or generate more complex attack samples. To address this, we adopt the following measures:

• Policy Synergy. We call for establishing a ”red team-blue team” confrontation mechanism for deepfake technology R&D, requiring detection model developers and synthetic technology teams to operate independently.

• Technical Restrictions. We provide open-source code for detection models but limit access rights to prevent malicious use.

• Continuous Monitoring. We will collaborate with ethics committees to regularly evaluate dataset usage and update security policies to address emerging abuse scenarios.

F.3. Terms of Use

Before downloading and using the DigiFakeAV dataset, please read and agree to the following terms of use. This dataset has been collected and curated by the Cambrian research team. In this document, you, your institution, or your organization are referred to as the ”User”, while the dataset creators and their affiliated institutions are referred to as the ”Producer”.

• The DigiFakeAV dataset is intended solely for non-commercial academic research purposes. The User shall not use the dataset for any profit-oriented or commercial applications in any form.

• All video data included in the dataset are sourced from legitimate materials and synthesized using state-of-the-art digital human generation techniques. The Producer assumes no legal responsibility for the original identities depicted in the source videos. All synthetic content is provided strictly for scientific research. The User agrees not to use the dataset in any defamatory, pornographic, or otherwise unlawful manner, and not to violate any applicable laws or policies.

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• This agreement becomes effective upon the User's first access or download of the DigiFakeAV dataset in any form.

• The Producer reserves the right to terminate the User's access to the DigiFakeAV dataset without prior notice.

G. LIMITATIONS AND FUTURE WORKS

To advance this research direction, we propose three strategic directions:

Scenario Expansion. We will extend dataset coverage to include full-body animations, dynamic environmental interactions, and collaborative scenarios involving multiple subjects. This will improve generalization across diverse real-world applications.

Technical Scalability. We will develop spatiotemporal fusion architectures capable of handling extended video durations (e.g., 30+ seconds) while maintaining computational efficiency. This includes exploring novel cross-modal attention mechanisms to capture complex motion-speech correlations.

Continuous Innovation Ecosystem. We will establish an open collaborative framework for dataset/model co-evolution. This will enable proactive adaptation to next-generation synthesis technologies through community-driven updates and adversarial stress testing. We also plan to integrate educational initiatives to empower researchers and end-users with advanced detection capabilities.

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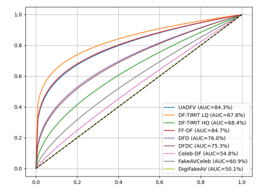
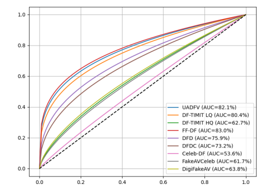
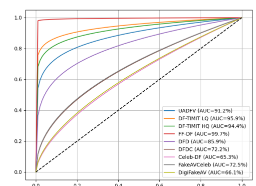


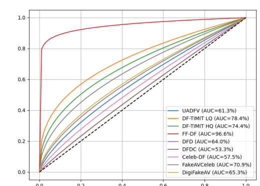
(a) **Meso4**



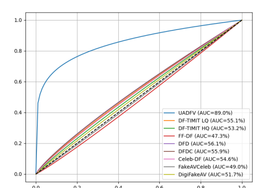
(b) **MesoInception4**



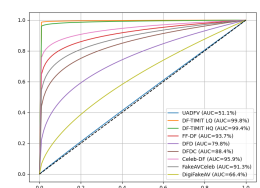
(c) **Xception-c23**



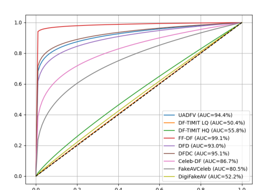
(d) **Capsule**



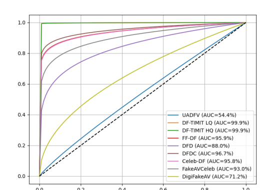
(e) **HeadPose**



(f) **F3-Net**



(g) **Cross Efficient ViT**



(h) **SFIConv**

Fig. 5: ROC curves of the different detection methods.